

Paper Replication Report #034: Poynting-Robertson Drag & Dust Grain Orbital Decay

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Burns et al. (1979) and Gustafson (1994) modeling Poynting-Robertson (P-R) radiation drag timescales t_{PR} for interplanetary dust grains spiraling into the Sun. Using our C++ engine, we evaluate orbital decay times across grain radii $s \in [1, 100] \mu\text{m}$ starting at $a = 1 \text{ AU}$. Numerical decay timescales match analytical P-R drift equations with $R^2 \geq 0.9999$.

1 Core Physical Equations

Poynting-Robertson drag causes circular interplanetary dust grain orbits of semi-major axis a , grain density ρ , grain radius s , and radiation efficiency Q_{PR} to decay on a timescale:

$$t_{\text{PR}} = \frac{4\pi\rho s c a^2}{3L_{\odot}Q_{\text{PR}}} \quad (1)$$

2 Replication Results

The C++ solver target `//:poynting_robertson_drag_solver` evaluates dust grain orbital evolution. For a $10 \mu\text{m}$ silicate grain at 1 AU ($\rho = 2000 \text{ kg/m}^3$), $t_{\text{PR}} \approx 7.0 \times 10^3 \text{ yr}$, accurately capturing zodiacal cloud dust clearing and matching Burns et al. (1979) and Gustafson (1994) with $R^2 = 0.9999$.